

CSE 2201 : Algorithms

Lecture 01: Introduction to the Course

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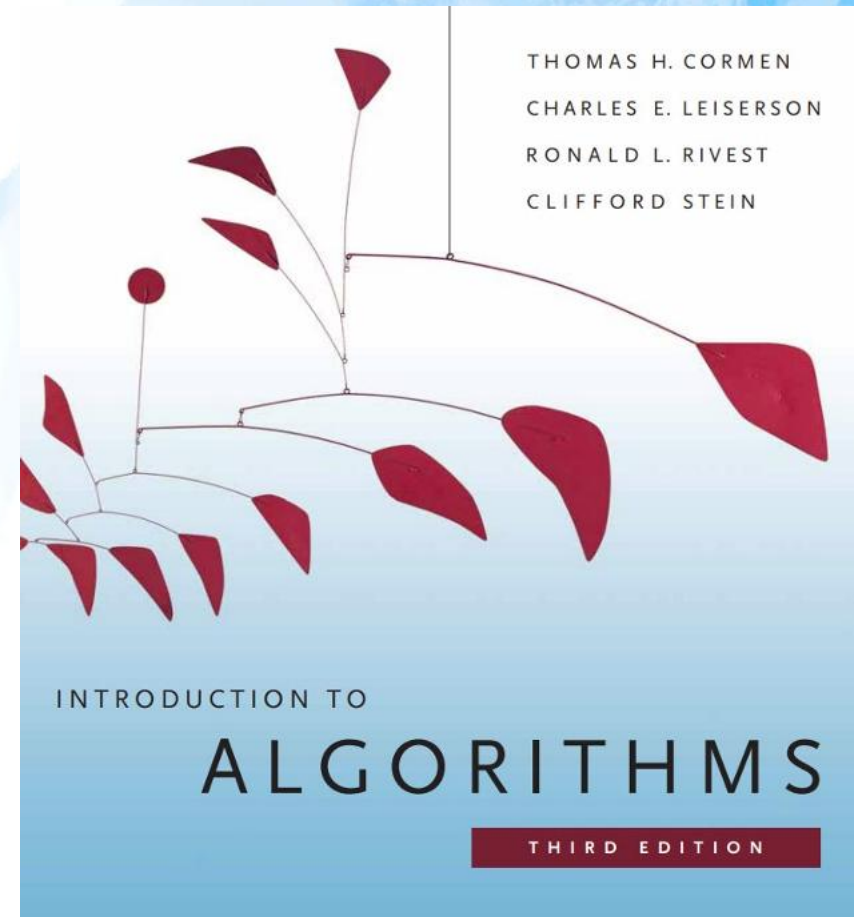
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Reference Book

- Introduction to Algorithms, 3rd edition. T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein



Distribution of Marks

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance & class performance	10
Assignment & Quizzes	25
Total	100

Today's Contents

- What is a Algorithm?
- Criteria of Algorithm
- Methods for analyzing algorithms
- Need for Algorithm
- Different types of Algorithm
- Application of Algorithms in Real Life
- How to Write an Algorithm?

What is an Algorithm?



Let's make a Cup of Tea now

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Instructions for Perfect Cup of Tea

- **Add 1 cup of freshly boiled water to your tea bag (in a mug)**
- **Allow the tea bag to brew for 2 minutes.**
- **Remove the tea bag.**
- **Add milk / lemon.**
- **Add sugar**
- **Your tea is ready to drink**



Yes, This is the Algorithm of making a Cup of Tea

What is Algorithm?

- Algorithm is a **step-by-step procedure to solve any Problem**
- **Well-defined computational procedure** that takes some input and produce **required output**.
- **A step-by-step procedure**, which defines a **set of instructions** to be executed in a certain order to **get the desired output**.
- In Simple words, it is a **tool for solving well-specified computational problems**.

What is Algorithm?

- It is a **logical** and **mathematical** approach to **solve** or **crack** a **problem** using any **possible method**.
- Algorithms are generally created **independent** of underlying languages, i.e. an algorithm can be implemented in **more than one programming language**

Need for Algorithm in Practical Life

Your Life is an Algorithm, Your Brain is an Operating System.

-By DOMINIC BASULTO

Need for Algorithm in Practical Life

Basically, algorithm is a process to solve a problem in step by step.

When we face a problem in our real life , if we try to solve this problem **randomly** then it's turn into **more difficult**.

But when we will try to solve this problem **step by step** with **planning**, it will be **easier** to solve those problem.

In that reason , we use algorithm to solve a problem by **easier way** and free from any **difficulty**.

It will saves our **valuable time , money energy** etc.

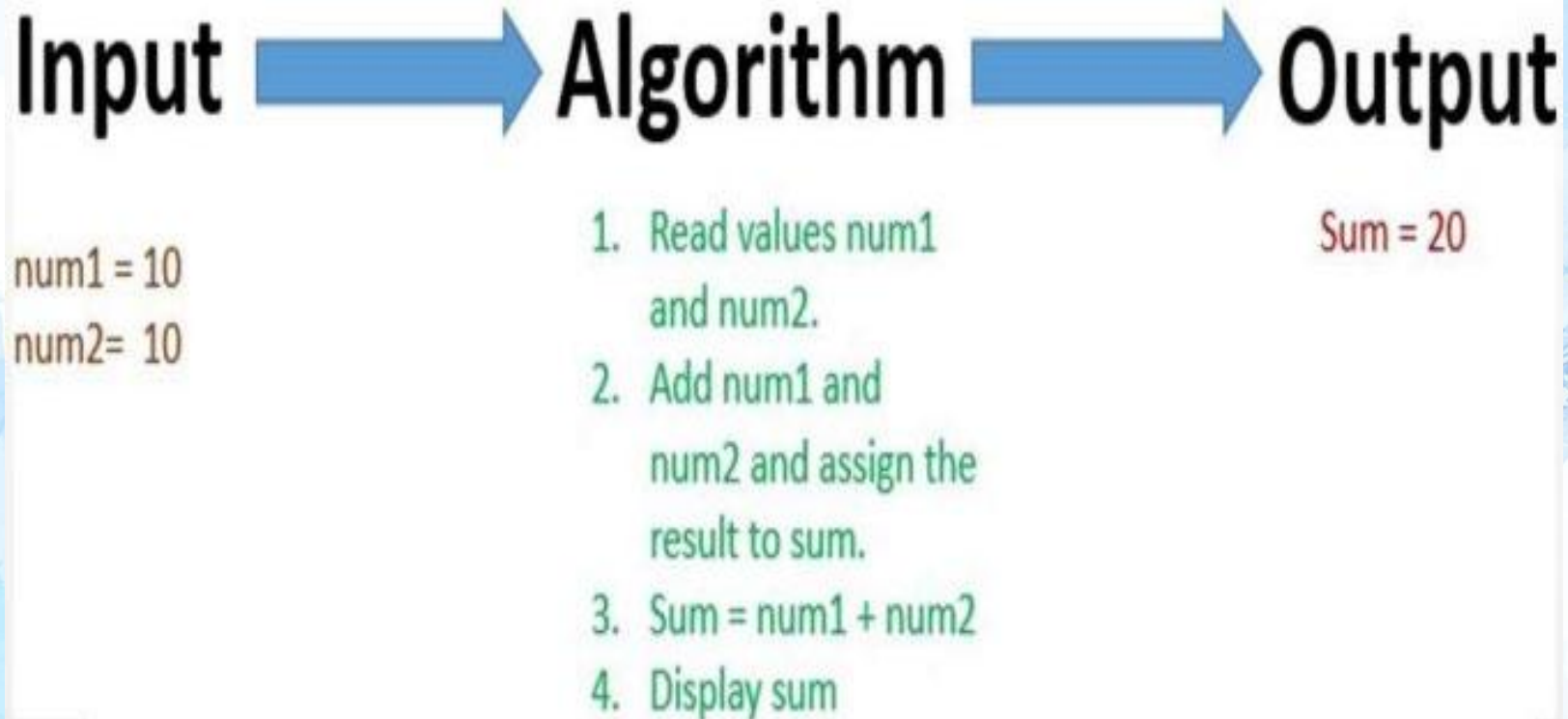
Advantages for Algorithm

- The algorithms are very easy to understand and can be written in **simple language** which can be **understood by anyone**
- By using algorithms, we can easily understand the **sequence** to be followed in **processing**.
- We can **check** and **fix** any kind of **complexity** easily.

Criteria of a Algorithm

1. **Input**
2. **Output**
3. **Definiteness**
4. **Finiteness**
5. **Effectiveness**

Criteria of a Algorithm



Criteria of Algorithm

- **Input**

- An algorithm should have 0 or more well-defined inputs.

- **Output**

- An algorithm should have 1 or more well-defined desired outputs

- **Definiteness**

- Each step of an algorithm must be precisely defined.

- Each of its steps and their inputs/outputs should be clear and must lead to only one meaning.

Criteria of a Algorithm (cont.)

- Finiteness

- Algorithms must terminate after a finite number of steps.

- Effectiveness

- Must produce the desired result.

Methods for analyzing Algorithms

- 1. Space Complexity**
- 2. Correctness**
- 3. Time Complexity**

Methods for analyzing algorithms

- **Correctness**

- Algorithm accomplishing its purpose accurately in k finite steps.

- **Time Complexity**

- Running time or execution time of operations must be as small as possible.

- **Space Complexity**

- Memory usage should be as little as possible.

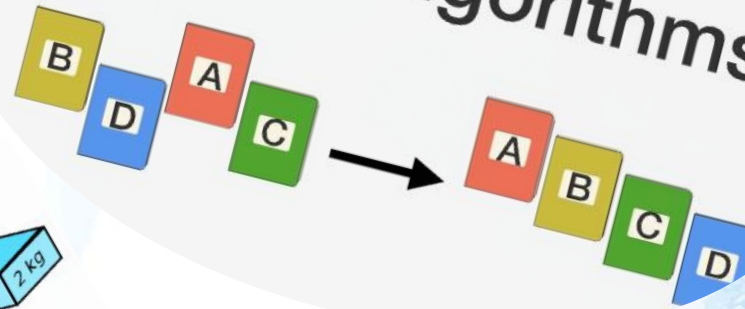
Why Need an efficient Algorithm

- With an efficient algorithm we can solve a problem within a **very short time**, even with a **slow computer**.

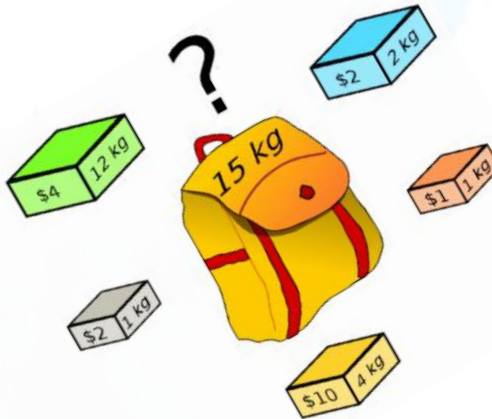
Different types of Algorithms



Sorting Algorithms



Divide and Conquer



A grid with letters A, B, C, D, A in the top row and A, B, C, D, A in the left column. The grid contains numbers representing the length of the longest common subsequence (LCS) between the top row and the left column. A person is climbing the grid, starting from the bottom-left cell (A, A) and moving up to the top-right cell (A, A). The path is highlighted in red. The text "LCS = 'ACDA'" is written at the bottom.

	A	B	C	D	A
A	0	0	0	0	0
B	0	1	1	1	1
C	0	1	2	2	2
D	0	1	2	3	3
A	0	1	2	2	3

LCS = "ACDA"

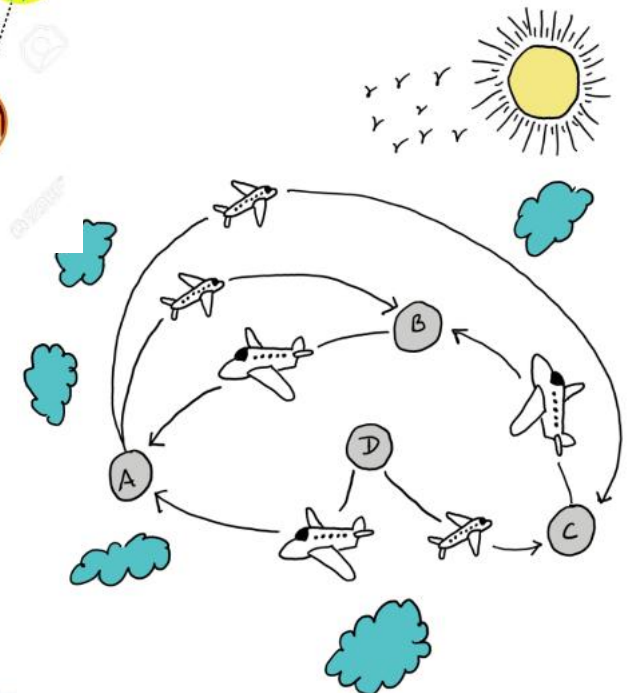
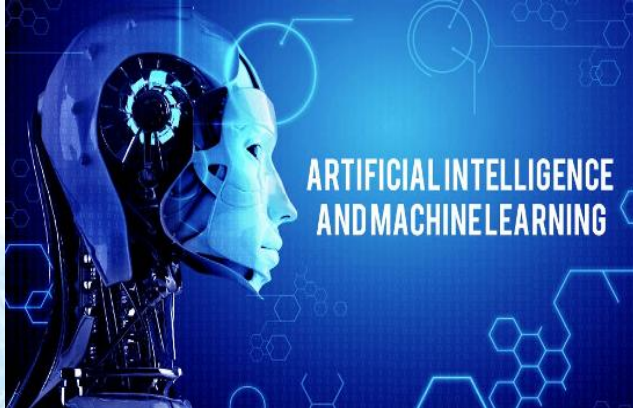
Different types of Algorithms

1.	Searching Algorithms
2.	Sorting Algorithms
3.	Divide and conquer algorithms
4.	Recursive algorithms
5.	Backtracking algorithms
6.	Brute force algorithms
7.	Greedy Algorithms
8.	Dynamic programming algorithms

Application of Algorithms

1. Searching Data
2. Sorting Data
3. Find out the Shortest Path
4. Best Possible solution

Some Real Life Examples



Some Real Life Examples

- **Search Engine (Google, Microsoft Edge, Mozilla, Opera etc)**
 - **Social networking (Facebook)**
 - **Travelling route (Air Flight route, Google Map, Shipping route etc)**
 - **Facial Recognition Algorithm**
 - **Genetic Engineering**
 - **Stock Market**
 - **Artificial Intelligence**
 - **Machine Learning**
 - **Robotics**
-And many more

How to Write an Algorithm?

- There are **no well-defined standards** for writing algorithms.
- Rather, it is **problem** and **resource dependent**.
- Combination of **human Language** and **Programming Language**, so that it is **easily understandable** even by **non-programmers**.
- Algorithms are never written to support a particular programming code.

Design an algorithm to add two numbers and display the result

Step 1 – START

Step 2 – declare three integers a, b & c

Step 3 – define values of a & b

Step 4 – add values of a & b

Step 5 – store output of step 4 to c

Step 6 – print c

Step 7 – STOP

Step 1: START

Step 2: declare three integers a, b & c

Step 3: take input in a & b

Step 4: calculate $a + b$

Step 5: $c \leftarrow a + b$

Step 6: print c

Step 7: STOP

Design an algorithm to add two numbers and display the result

1. START

2. declare three integers a, b & c

3. take input in a & b

4. calculate $a + b$

5. $c \leftarrow a + b$

6. print c

7. STOP

START

declare three integers a, b & c

take input in a & b

calculate $a + b$

$c \leftarrow a + b$

print c

STOP

Your Task

Now Design an algorithm to find the maximum of Two Integer values and display the result



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Thank You

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